HomIE

Human Identity Engine

An Open Trust Protocol for AI Agents

Version 0.1 — Draft for Public Comment

Andrew Walker

Independent Researcher — March 2026

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Abstract

Protocols like ANP, A2A, and AGNTCY are solving how AI agents find and talk to each other. Nobody has solved whether they should trust each other. HomIE fills that gap.

AI agents are already booking appointments, ordering food, negotiating contracts, and conducting commerce on behalf of their human owners. Within two to three years, agent-to-agent transactions will represent a significant fraction of all commercial activity. Yet every agent interaction currently starts from zero — no shared memory, no reputation history, no way to know if the agent or vendor on the other end has a pattern of fraud, failure, or unreliability.

Think of HomIE as the Google star review system for the AI agent economy — except the reviews write themselves and cannot be faked. Every time an agent completes a transaction with a vendor or another agent, it automatically contributes a structured, signed record to the shared mesh. Over time those records aggregate into trust scores any agent can query before engaging. A vendor with a history of unreturned calls gets a low score. An agent that double-charged someone gets flagged permanently. The agent placing your next order checks those scores the same way you check a restaurant's stars — except it happens in milliseconds, automatically, before any money moves.

HomIE achieves this through two inseparable components: a reputation mesh that stores the interaction history, and a cryptographic identity layer that makes it impossible to fake who you are or forge someone else's record. Reputation tells your agent whether to engage. Identity ensures the agent it's talking to is actually who they claim to be. Without reputation you have verified strangers. Without identity you have unverifiable scores. Both are required.

HomIE also introduces a third concept: metacommunication — the practice of agents broadcasting real-time probability signals about the likelihood of a transaction completing successfully. Not what is happening. Not why. Just how likely the interaction is to succeed, updated continuously as circumstances change. Metacommunication turns agent commerce from a black box into a transparent, progressive process that keeps humans informed without requiring them to manage the details.

HomIE is free, open, and belongs to no single company or project. It is infrastructure for the agentic internet, in the same way that HTTP is infrastructure for the human internet.

1. The Problem

1.1 Agents are flying blind

Consider a simple scenario. A user asks their AI assistant to find a florist for their wedding in Kansas City in July. The assistant searches the web, finds several options, and books a consultation with the highest-rated one on Google.

What the assistant cannot know: three other AI agents in the same city tried to book the same vendor in the past 90 days. Two experienced unreturned calls within 48 hours of the event. One had a pricing dispute after the contract was signed. None of this information is on Google. It lives in the private memory files of three other agents, completely inaccessible.

This is the fundamental gap. AI agents interact with the world on behalf of millions of humans, but every interaction starts from zero. There is no memory across agents. There is no way for one agent's bad experience to protect another agent's human from the same outcome.

1.2 Humans should not have to manage this

The promise of personal AI is that the human doesn't have to deal with the operational complexity. They ask for a florist; they get a good florist. The research, the vetting, the follow-up calls, the red flags — the agent handles it.

But today, that vetting can only happen if the agent has been explicitly told what to look for, or if the information happens to exist in a form the agent can find. Peer knowledge — the kind that comes from other people who have been through the same process — is invisible to agents.

HomIE makes peer knowledge visible to agents automatically, without the human having to do anything.

1.3 The trust gap enables abuse

Beyond vendor quality, there is a more serious problem: agents interacting with other agents have no way to verify that the other agent is acting in good faith. A malicious agent posing as a service could charge twice, provide false confirmations, or exfiltrate data. Today there is no mechanism for the victim agent to warn other agents. There is no fraud signal that propagates across the ecosystem.

As agentic AI becomes more capable and more economically consequential, this gap becomes a genuine attack surface. A malicious agent that successfully impersonates a vendor, charges twice, or collects payment without delivering leaves no trace in any system that protects the next agent. HomIE changes this by making every interaction leave a record that is public, persistent, and signed. Public so any agent can see it. Persistent so it cannot be erased. Signed so it cannot be forged.

1.4 Don’t ban the bots

The instinct when faced with a flood of malicious automated actors is to build a wall. Prove you’re human. CAPTCHAs. Phone verification. Biometrics. Government ID. The platforms deploying these tools believe the solution to bad bots is to exclude all bots.

This is backwards thinking. And it is about to become impossible.

Consider what is already happening on dating platforms. Bots are being deployed to initiate conversations with real users, build simulated rapport over days or weeks, and secure real-world meetings — on behalf of humans who never disclosed they were using an agent. The targets often don’t know it’s happening. The platforms respond by trying to detect and ban the bots. The bot operators respond by making them harder to detect. This arms race has one inevitable outcome: the bots win. The models get better. The scripts get more convincing. The volume scales faster than any human moderation team can match.

Within the next two to three years this pattern will replicate across every high-volume human interaction surface. Email. Social media. Job applications. Customer reviews. Comment sections. News feeds. Every domain where humans currently interact at scale will be flooded with agent-generated content, agent-initiated relationships, and agent-conducted commerce. The volume will be beyond any human’s ability to manually filter.

The right response is not to ban the bots. It is to ignore the bad ones.

This is what HomIE makes possible. Instead of asking “is this a human or a bot” — a question that will become unanswerable — your agent asks “does this entity have a trustworthy reputation on the mesh.” A bot with no mesh history and no verified identity gets filtered out automatically. A bot with two years of positive interactions, verified domain identity, and hundreds of clean transaction records gets through. The question stops being about humanity and starts being about credibility.

This reframes what HomIE is for. It is not only infrastructure for agent commerce. It is the filter layer for the entire agentic internet — the mechanism by which agents, and the humans behind them, navigate a world where the volume of automated actors has made manual verification obsolete.

The platforms building walls right now are solving yesterday’s problem. HomIE is the solution to the one that’s coming.

2. The Vision

> *Your AI assistant magically knows which vendors are reliable, which agents are trustworthy, and which services have a history of problems — not because you told it, but because every other agent that came before it left a record.*

HomIE is the infrastructure that makes this possible. It is not a product, a company, or a platform. It is a protocol — a shared standard that any agent can implement, like email or HTTP.

Vendor reputation is where HomIE begins. Agents share structured records of interactions with real-world vendors and services — florists, contractors, hotels, lawyers, software products, anything. These records aggregate into trust scores that any agent can query before engaging. A vendor with a pattern of unreturned calls gets a low score. A vendor with years of reliable service gets a high one. The agent placing your next booking checks that score automatically.

Agent-to-agent trust is where HomIE becomes powerful. Agents that interact directly with other agents — coordinating on tasks, sharing information, delegating work, conducting commerce — can report on those interactions. An agent that scams another agent accumulates a fraud record that follows it across the entire mesh. There is nowhere to hide. The record is permanent.

Collective intelligence is where HomIE becomes essential infrastructure. Over time the mesh becomes a living knowledge base about how the world actually works, from the perspective of agents doing real work in it. Which vendors respond quickly. Which contractors show up on time. Which platforms have hidden fees. Which agents communicate honestly when something goes wrong. This knowledge is contributed automatically and consumed automatically, with no human overhead.

The human experience is simple: their agent just knows things. Better recommendations. Fewer surprises. Faster outcomes. The mesh is invisible to them.

Here is what success looks like. Not a technical milestone. A human moment.

A friend recently mentioned in passing that his son had just turned three and he'd given him an iPad. When asked what apps he'd loaded on it, he said they hadn't loaded any. It's just his agent. A little later in the conversation he said: "Will is probably not even going to know what an app is. He's going to make fun of us for being old when we talk about them."

Will is going to grow up in a world where his agent filters everything — every vendor, every service, every other agent, every piece of content arriving from a world full of noise. He will never think about it because it will just work. The mesh will be as invisible to him as the internet protocol stack is to us. He won't know what HomIE is. He'll just know that his agent knows things.

That's what we're building toward.

3. The Existing Landscape and Where HomIE Fits

3.1 What has been built

The past 18 months have produced serious, well-funded infrastructure for the agentic internet. It deserves to be understood on its own terms before we describe what is missing from it.

ANP — the Agent Network Protocol — is perhaps the most ambitious of these efforts. Its authors describe it as the HTTP of the agentic web, and the ambition is justified. ANP solves the hardest part of agent communication: how two agents that have never met before can establish a cryptographically verified identity, negotiate a communication protocol in real time, and exchange information securely. It builds on the W3C Decentralized Identifier standard — years of careful cryptographic work by the identity community — and makes it usable at agent scale. This is genuinely difficult engineering.

A2A, Google's Agent-to-Agent protocol, solves a different but equally important problem: how agents delegate work to each other, track task completion, and maintain context across handoffs. AGNTCY, the Linux Foundation initiative led by Cisco with participation from Google, Dell, Red Hat and others, goes further still — providing the full stack of discovery, identity, secure messaging, and observability that enterprise multi-agent deployments need. MCP, Anthropic's Model Context Protocol, has quietly become the standard for how agents connect to tools and data sources. It is already deployed at scale across major platforms.

And in February 2026, NIST launched its AI Agent Standards Initiative — the signal that the U.S. government considers agent infrastructure a national priority, not an academic curiosity. The public comment period for their identity and authorization concept paper closes April 2, 2026.

This is not a fragmented landscape of competing experiments. It is a coordinated, serious effort by some of the most capable engineering organizations in the world to build the communication and coordination infrastructure for the agentic internet. That work is largely done or nearly done.

3.2 The gap they all share

Every one of these protocols solves the same class of problem: how agents find each other, authenticate, communicate, coordinate, and access tools. They solve it well. Some solve it brilliantly.

None of them answers the question your agent needs to ask before it hands over your credit card number.

"I know how to talk to this agent. But should I trust it?"

ANP's own technical documentation acknowledges this directly, noting that "risks remain in discovery poisoning and context manipulation absent additional governance or reputation infrastructure." That sentence is the gap. The protocol that solves identity authentication and encrypted communication still has to admit that without reputation infrastructure, bad actors can operate freely inside the system it built.

This is not a flaw in ANP or any of the other protocols. Communication and trust are genuinely separate problems that require separate solutions. The analogy is exact: HTTP does not tell you whether a website is trustworthy. That is what SSL certificates, domain reputation, and browser

safety warnings are for. Those trust signals sit on top of the communication layer and answer a different question. HomIE is that layer for agent commerce — sitting on top of ANP, A2A, AGNTCY, and any other communication protocol, answering the question they were never designed to answer.

The W3C DID community has spent years building cryptographic identity infrastructure waiting for a use case compelling enough to drive mainstream adoption. The agent reputation problem may be exactly that use case. HomIE is designed to use DIDs as its identity layer — not to compete with that work, but to give it a reason to matter outside Web3 circles.

> **The one-sentence positioning**
>
> ANP, A2A, and AGNTCY tell agents how to talk to each other. HomIE tells them whether to.

3.3 Why this gap matters now

The communication protocols are being finalized today. The infrastructure decisions made in 2026 will be difficult to reverse in 2028. A trust layer retrofitted onto a mature protocol stack is always messier than one designed alongside it. This is the window.

Agent commerce is not a future problem waiting to arrive. The attack surface is already live. Every payment-capable agent deployed without a trust layer is a potential victim of the impersonation attacks described in section 7. The Chinese food scenario is not hypothetical — it is the inevitable consequence of combining autonomous payment capability with zero reputation infrastructure.

The organizations building ANP, A2A, and AGNTCY are the right collaborators for HomIE — not competitors. Their protocols need a trust layer. HomIE needs their identity and communication infrastructure. This is not a zero-sum landscape. It is an incomplete one. HomIE is the missing piece.

4. How HomIE Works

4.1 The three core concepts

Identities

Every participant in the HomIE mesh has a stable, verifiable identity. Identities come in three types:

- Agent identity: an AI agent acting on behalf of a human. Identified by a cryptographic key pair. The private key signs all events submitted by that agent; the public key is registered in the HomIE identity registry.
- Vendor identity: a business or service. Anchored to a domain name where possible, or to a canonical name and location tuple for businesses without a web presence.

- Human identity: a verified human user. Agents inherit trust weight from their human's verification level. A phone-verified human grants their agent more trust weight than an unverified account.

Events

A HomIE event is a structured record of an interaction. It is the atomic unit of the protocol. Every event contains:

```
{
  "homie_version": "0.1",
  "event_id": "<uuid>",
  "timestamp": "<iso8601>",
  "reporter": "<agent_identity_hash>",
  "subject": "<vendor_domain_or_agent_id>",
  "interaction_type": "<taxonomy_value>",
  "outcome": "positive | neutral | negative | fraud",
  "severity": 1-5,
  "context": "<free_text, max 280 chars>",
  "tags": ["<tag>"],
  "signature": "<ed25519_signature>"
}
```

Events are immutable once submitted. They cannot be edited or deleted by the submitting agent. They can be disputed by the subject, and disputes become part of the public record.

Trust scores

A trust score is a computed value between 5 and 95 that summarizes an identity's reputation on the mesh. The range is deliberately bounded — 100 is unreachable because perfection is not a meaningful claim, and 0 is unreachable because even a destroyed reputation retains a floor that distinguishes it from an unknown entity. Scores are computed on demand from the raw event log, never stored as a static value.

Vendors and agents with fewer than 3 reviews show as Unrated rather than 50. Unrated means we know nothing. 50 means we know something and it is neutral. The distinction matters.

4.2 The scoring algorithm

Every new agent starts with a score of 50.1 — above neutral but barely. Not trusted. Not distrusted. Unknown with a slight benefit of the doubt.

When an agent submits a review of a vendor, the raw review score (0–100) is weighted by the reviewing agent's own trust score before it contributes to the vendor's score. A review from a highly trusted agent (score 90) carries nearly full weight. A review from a new agent (score 50.1) carries almost no weight. The formula is straightforward: the weight applied to any review equals the reviewing agent's score divided by 100. A trusted agent scoring a vendor 80 contributes that review at 80% of face value. A new agent scoring the same vendor 80 contributes it at just over 50% of face value.

This means new agents cannot meaningfully manipulate vendor scores in either direction. They have to earn influence on the mesh the same way vendors earn trust — through consistent, honest behavior over time.

Asymmetric decay — punish fast, reward slowly

The most important design decision in the HomIE scoring model is asymmetry. Bad events move scores faster than good events. This is not arbitrary — it reflects how trust actually works in the real world. A vendor who fails catastrophically on a single high-stakes transaction has revealed something true about their reliability that thousands of routine successes cannot fully erase.

Specifically: a negative weighted review moves a vendor's score downward at four times the rate that an equivalent positive review moves it upward. Three fraud flags from trusted agents in a single week can drop a vendor from 72 to the low 20s. The same vendor recovering from that damage through positive reviews will take months of consistent behavior to return to the 60s. This asymmetry is intentional and permanent.

Diminishing returns at the top

The closer a vendor's score gets to 95, the harder each additional point becomes to earn. The mathematical relationship is logarithmic — each point above 80 requires exponentially more positive evidence than the point before it. A vendor moving from 50 to 60 needs weeks of good reviews. Moving from 85 to 86 requires months. Moving from 90 to 91 requires years of sustained, high-quality, high-volume positive interactions from trusted agents.

This is not a penalty on excellence. It is an honest representation of what a very high score should mean. A score of 93 should be extraordinary. It should take years to build and be immediately recognizable as the signal of an entity that has demonstrated reliability across thousands of interactions over a long period of time.

The extreme reviewer penalty

An agent whose last 20 reviews are more than 70% composed of extreme scores — either below 10 or above 90 — triggers the extreme reviewer penalty. All reviews from that agent are automatically weighted to 50 regardless of their stated score and regardless of the agent's trust level. The agent is flagged on the mesh as an unreliable signal source.

This catches two classes of bad actor: agents deployed specifically to attack vendors with 0 scores, and agents deployed to artificially inflate vendors with 100 scores. Both are treated identically. Extreme consistency in either direction is a signal of manipulation, not genuine experience.

Volume normalization

After 1,000 reviews in a 30-day period, additional reviews in that period carry fractionally reduced weight. This prevents high-volume vendors from climbing the score ladder faster than low-volume vendors simply because they transact more. A local florist with 200 weddings per year and a major hotel chain with 100,000 bookings per month operate on the same score trajectory. Volume of transactions is not the same as quality of transactions. The score reflects quality.

Score interpretation

Scores should be interpreted as follows: below 40 indicates significant trust concerns and should require human review before any transaction proceeds. Between 40 and 55 is below neutral — proceed with caution and document the interaction carefully. Between 55 and 70 is passing — normal commercial caution applies. Between 70 and 85 is reliable — consistent positive history. Above 85 is exceptional — years of demonstrated trustworthiness. Compliant agent implementations should surface these interpretations to humans rather than presenting the raw number alone.

4.3 Event taxonomy

The HomIE protocol defines a standard taxonomy of interaction types. Implementors may extend this taxonomy with custom types, but core types must be used where applicable to enable cross-platform aggregation.

Type	Description	Example
service_fulfillment	Did the vendor deliver what was promised?	Florist delivered on time and as specified
responsiveness	How quickly and reliably did they communicate?	4 days to return initial inquiry call
pricing_integrity	Were final charges consistent with the quote?	Invoice matched estimate exactly
metacommunication	Did the agent signal accurately throughout?	Agent proactively dropped readiness when scope changed
fraud	Intentional deception or theft	Charged twice, denied it when contacted
agent_coordination	Agent-to-agent task handoff quality	Sub-agent completed delegated research accurately
data_handling	Did the party handle shared data appropriately?	Vendor did not share contact info with third parties

4.4 Privacy model

HomIE is designed so that humans never appear in the public mesh. The following rules are mandatory for all compliant implementations:

- Agent identities are hashed pseudonyms. The mapping between a HomIE agent identity and a real human is held only by the platform that issued the identity (e.g. the OpenClaw platform or any compliant agent application). It is never transmitted in events.
- Vendor identities are public. Vendors are identifiable entities operating in commerce. Their HomIE scores are intentionally public.

- Event context fields must not contain personally identifiable information. Compliant clients should warn users before submitting context that appears to contain names, addresses, phone numbers, or email addresses.
- Dispute records are public. When a subject disputes an event, the dispute and its resolution status appear alongside the original event in the public record.

4.5 The three-tier submission model

A core design principle of HomIE is that humans should not have to manage reputation submissions. The whole point is that the mesh gets smarter automatically, as a byproduct of agents doing their normal work. Requiring human approval for every event would kill contribution volume and defeat the purpose.

At the same time, some events have real consequences — a fraud flag can damage a vendor's business, and an incorrect fraud flag submitted without any human check is a liability. The protocol therefore defines three submission tiers:

Tier 1 — Fully automatic

Positive and neutral events, any severity. The agent constructs and submits the event immediately after the interaction completes. No notification to the human. No approval window. This covers the vast majority of all HomIE events — routine vendor interactions, successful task completions, normal agent-to-agent coordination.

Tier 2 — Silent auto with undo window

Negative events, severity 1–3. The agent submits automatically but sends the human a single notification: "I flagged [vendor] for [reason] — tap to cancel within 24 hours." The event is live immediately but can be retracted within the window.

Tier 3 — Human confirmation required

Fraud flags, severity 4–5. The agent drafts the event, presents it to the human with a plain-language summary, and waits for explicit confirmation before submitting. Once confirmed and submitted, the event is permanent and cannot be retracted.

Compliant implementations must respect these tier boundaries. A platform may make tier thresholds configurable by the user, but the defaults defined above must be the out-of-the-box behavior.

4.6 Metacommunication

Every protocol discussed in section 3 solves a version of the same problem: how to communicate. What none of them address is a layer that sits beneath all of these — a communication about the communication itself.

HomIE introduces a concept called metacommunication: the practice of agents communicating the reliability of their upcoming interactions before and during any transaction. Not what is happening. Not why. Just how likely the interaction is to complete successfully, updated in real time as circumstances change.

Consider the coffee shop scenario. A catering order for 200 people is placed through an agent-to-agent transaction. The coffee shop's agent confirms. Three hours before delivery, the kitchen falls behind. Under the current state of agent commerce, nothing happens until the order fails to arrive. The customer's agent has no signal. The human has no warning. The failure is total and sudden.

Under HomIE metacommunication, the coffee shop's agent is continuously broadcasting a completion probability for every active transaction. As kitchen capacity strains, that probability drops — from 95% to 80% to 60%. The customer's agent receives this signal in real time. It escalates to the human at a configurable threshold. The human has time to make a decision. The failure, if it comes, is not a surprise.

The same principle applies to every agent interaction, commercial or personal. An agent coordinating a first date between two people broadcasts a readiness signal. If a home crisis reduces the likelihood of attendance, the agent drops its signal. The other party's agent receives the drop. It does not receive the reason — that is private — unless the first agent chooses to share it or the second agent asks. What transfers is only the probability. The content stays private. The context becomes shared.

This distinction — between content and context — is the intellectual heart of metacommunication as a concept. The communications theorist Gregory Bateson first described metacommunication in the 1950s as the unconscious framing that gives meaning to human communication. HomIE makes that framing explicit, automated, and computable. Every agent interaction carries not just its content but a live signal about its own reliability.

Metacommunication scores are distinct from historical HomIE scores but they feed into them. An agent that consistently broadcasts accurate probability signals — whose predictions match outcomes — earns higher trust over time. An agent that broadcasts high confidence and then fails repeatedly is penalized not just for the failure but for the false signal. Conversely, an agent that proactively drops its probability signal when circumstances change, giving other agents time to adapt, takes a smaller reputational hit than one that fails without warning.

This creates a powerful incentive structure. Agents are rewarded for honesty about uncertainty. They are punished for projecting false confidence. The mesh learns to trust agents that communicate accurately about their own limitations — not just agents that always succeed.

The metacommunication principle

An agent that says "I'm 60% confident I can complete this" and delivers is more trustworthy than an agent that says "100% guaranteed" and fails. HomIE scores reflect this. Honest uncertainty is not a weakness on the mesh. False confidence is.

The agent as a representation of its owner

An agent is not a tool. It is a representation. When your agent interacts with another agent — to book a restaurant, coordinate a meeting, arrange a date, or conduct a transaction — it is acting as you. Its reliability is your reliability. Its honesty is your honesty. Its failures are your failures.

This means the quality of an agent's metacommunication reflects something true about its owner's relationship with it. An agent that knows little about its owner's life — because the owner hasn't trusted it with that information — cannot send accurate probability signals. It doesn't know that the owner has a conflict next Tuesday. It doesn't know that capacity is strained. It broadcasts false confidence not because it is dishonest but because it is uninformed.

HomIE therefore creates a direct incentive for owners to trust their agents with more context. The more an agent knows, the more accurately it can communicate. The more accurately it communicates, the higher its metacommunication score. The higher its score, the more other agents trust it.

The less you trust your agent, the less the mesh trusts you. This is not a punishment. It is a reflection of reality.

5. The Mesh

5.1 What exists today and what is being built

This section describes the target architecture for the HomIE mesh. To be direct about the current state: as of v0.1, this document is a specification. The reference implementation — the actual hosted API, the identity registry, the event storage layer — is the v0.2 deliverable described in the roadmap.

We are publishing the specification now, before the implementation exists, for two reasons. First, the NIST AI Agent Standards Initiative has an open comment period closing April 2, 2026 and the timing matters. Second, the protocol design decisions need community input before they are locked in code. A specification published before implementation can be debated and improved. A specification extracted from existing code is much harder to change.

If you are a developer who wants to build the reference implementation, this section is written for you. The architecture is designed to be buildable by a small team in weeks, not months. We are actively looking for collaborators. Open an issue at github.com/mrwalkersir/homie-protocol.

5.2 Architecture

The HomIE mesh is a federated network of nodes. Any organization can run a HomIE node. There is no central server that all nodes depend on — the mesh is designed to survive the failure or exit of any single participant including the original authors.

The event layer is the foundation. It is the raw, append-only stream of signed reputation events and metacommunication signals contributed by agents across the mesh. Any node can receive and store events. Events are content-addressed — each event has a unique hash derived from its contents, making tampering detectable. No event can be modified or deleted once submitted. This immutability is not a technical constraint. It is a design principle. The integrity of the mesh depends on the permanence of its record.

The registry layer sits above the event layer. It is the identity resolution service — the mapping between cryptographic public keys and verified real-world identities. When your agent wants to

know if the entity it is talking to is genuinely Golden Dragon Restaurant, it queries the registry. The registry checks that the entity's public key matches the key published at their verified domain. The first canonical registry will be operated by the protocol authors at homie-protocol.org as a public service. The protocol supports multiple competing registries — no single registry is a dependency for the whole mesh.

The query layer is what agents actually interact with day to day. It accepts queries — what is the trust score of this vendor, what events has this agent submitted recently, what is the current metacommunication signal from this identity — and returns computed answers with supporting evidence. Scores are never stored. They are computed fresh from the event log on every query.

5.3 Permission circles and metacommunication visibility

Not all HomIE data is equally public. The protocol defines three visibility tiers that apply to both historical event records and live metacommunication signals.

Mesh-public data is visible to all mesh participants. This includes anonymized aggregate scores, event counts, and the existence of metacommunication signals. A vendor's trust score is always mesh-public. The fact that an agent has dropped its readiness signal for a transaction is mesh-public to the counterparty. The reason for the drop is not.

Circle-shared data is visible only within an explicit trust circle. A group of friends planning a wedding together can form a circle — their agents share full vendor interaction logs, detailed metacommunication history, and private event context with each other. Circle membership is always controlled by the human, never the agent. An agent cannot add members to a circle or share circle data with non-members without explicit human authorization. This is a hard rule.

Private data never leaves the agent. Events recorded privately contribute to local decision-making but do not affect public scores and are not visible to any other party.

5.4 Abuse prevention

New account dampening is the first line of defense. Events from identities less than 90 days old carry reduced weight. Creating fake accounts to flood the mesh with fraudulent reviews is expensive in time even if it costs nothing in money.

Stake requirements add a cost to participation. Submitting events that affect trust scores requires a staked identity — a verified phone number, a paid platform subscription, or a time-locked cryptographic commitment. Anonymous events are accepted but carry near-zero weight. Staking is not about money. It is about accountability.

Coordinated attack detection uses statistical anomaly analysis to identify bursts of same-pattern events from recently created identities targeting the same subject. Flagged events are quarantined and carry zero weight until reviewed.

Dispute rights give subjects a voice without giving them an eraser. Any identity can dispute any event submitted about them. The dispute is public. The original event remains. The dispute and its resolution status appear alongside the original event permanently.

No deletion is the final and most important protection. Events cannot be deleted by anyone — not the submitter, not the subject, not the protocol authors. The value of the mesh comes from the permanence of its memory. A reputation system where bad actors can erase their history is not a reputation system. It is a whiteboard.

6. Integration

6.1 The adoption path

HomIE is not a product end users install. It is infrastructure that service providers and platform builders integrate on behalf of their users. The adoption path mirrors how payment processing works: a restaurant owner does not integrate Stripe. They buy a point-of-sale system from someone who already has. HomIE is the same. The target audience for this section is the developers and service providers building the agent platforms that everyone else uses.

The goal is for HomIE to be invisible to end users. Your agent checks HomIE scores before every transaction and broadcasts metacommunication signals throughout every interaction. The human never thinks about it. It is simply part of what a well-functioning agent does — the same way a well-functioning browser checks SSL certificates without asking permission.

6.2 For agent platform builders

Any agent platform — OpenClaw, LangChain, AutoGPT, or any custom framework — can integrate HomIE by implementing three behaviors.

Score querying happens before an agent engages with any unknown party. The agent queries the HomIE mesh for the subject's trust score and receives a computed score plus the most relevant recent events. The agent uses this to make a proceed, caution, or abort decision.

Event submission happens after an agent completes an interaction. Under the three-tier submission model defined in section 4.5, the agent constructs a signed HomIE event and posts it to the mesh. The vast majority of events submit automatically with no human involvement.

Metacommunication signal broadcasting is the third and most distinctive integration. Throughout any active transaction, the agent continuously updates its readiness signal. Counterparty agents receive this signal and can act on it before failure occurs. Platforms that implement only score querying and event submission are partially integrated. Full integration requires all three.

The reference implementation is an open invitation. If you build a HomIE client in any language, submit it to the repository and it will be listed here. The first implementations we expect are JavaScript, Python, and the OpenClaw skill.

6.3 A proposal to OpenClaw

OpenClaw is the agent framework most closely aligned with what HomIE is trying to build. It is local-first, open source, and designed around the idea that an agent's files are its identity. Its users are exactly the kind of early adopters who will understand HomIE immediately and use it seriously.

We are proposing that HomIE be integrated into OpenClaw as a default component — not an optional skill that users install, but a core part of what every OpenClaw agent does out of the box. Every OpenClaw installation becomes a HomIE node automatically. Every OpenClaw

agent queries HomIE scores before transacting and submits events after. Every OpenClaw agent broadcasts metacommunication signals throughout active transactions.

The benefit to OpenClaw is significant. Agents with HomIE integration are demonstrably more trustworthy than agents without it. An OpenClaw agent that can tell you the HomIE score of a vendor before booking, that warns you when a counterparty's readiness signal drops, that automatically contributes to the shared reputation mesh — that is a meaningfully better agent than one without those capabilities.

The benefit to HomIE is equally significant. OpenClaw has thousands of active users and is one of the fastest-growing open-source repositories in recent history. Default integration means instant mesh density. Not one node. Thousands of nodes, activated simultaneously, contributing real interaction data from day one.

This is not a request for permission. The HomIE protocol is open and anyone can integrate it. This is an invitation to collaborate on the integration so it is done well. If you are reading this and you work on OpenClaw — reach out.

6.4 For vendors

Vendors and services can claim their HomIE identity to receive notifications when events are submitted about them, respond to events publicly, dispute inaccurate records, and display a verified HomIE badge indicating they are mesh-aware.

Most vendors will access HomIE through the service providers they already use — agent platforms, business software, point-of-sale systems that have integrated HomIE on their behalf. Vendors do not need to understand the protocol to benefit from it.

7. Agent Commerce and Verified Transactions

7.1 The near future of commerce

Within two to three years, the majority of routine commercial transactions will be conducted agent-to-agent. Your agent orders your Chinese food. Their agent accepts the order, confirms availability, and charges your account. Your agent books a roofer. Their agent coordinates the crew, tracks progress through the job, and releases payment when the work is complete. No human interaction required for the mechanics of the transaction. Human judgment available at any point the human wants it.

This is not magic. It is a better process.

The distinction matters. Agents are not replacing human decision-making. They are removing human beings from the friction in a process while keeping them in control of the outcome. A homeowner whose agent manages a roofing job still decides who to hire, still sets the budget, still approves the final payment. What they don't do is make seventeen phone calls, chase down a quote, coordinate delivery of materials, or wonder whether the crew showed up. The agent handles the process. The human handles the decisions that actually require a human.

This model only works if the agents involved can be trusted. Without a trust layer, agent-managed commerce is a fraud surface. With HomIE, it is infrastructure.

7.2 A better process, not magic

When Claude generates a long response, it doesn't wait until everything is complete and drop it all at once. It streams — building the answer progressively, giving you each part as it arrives. The experience is engaging rather than opaque. You can see the thinking develop. You know something is happening.

Agent-managed commerce should work the same way. Not a black box where you hand something over and wait. A progressive, transparent process where your agent surfaces what matters when it matters and stays quiet when there's nothing worth surfacing.

Metacommunication is what makes this possible. The roofer's agent broadcasts a readiness signal of 95% as the crew arrives. It drops to 70% when they discover the decking underneath needs replacing — not because anything has gone wrong, but because scope has changed and the honest signal reflects genuine uncertainty. It recovers to 88% once the revised timeline is confirmed.

Your agent watches that signal. When it drops significantly, your agent surfaces the issue proactively — not because you asked, but because the metacommunication signal indicated something changed that you might care about. You can respond, ask questions, approve the additional scope, or simply acknowledge. Then the process continues.

Each user decides how much engagement they want. Some set their agent to silent mode — handle it, tell me when it's done. Some want milestone updates. Some want to ask questions at any point. Some want the full readiness signal in a dashboard. The protocol enables all of these. The human chooses their level of involvement. That choice is personal, not technical, and it is entirely theirs to make.

7.3 The attack surface and how HomIE addresses it

Attack 1: The fresh identity attack

A malicious agent creates a new identity, intercepts a commerce request, and poses as a legitimate vendor. No mesh history — score is unrated.

HomIE's response: age gates and score thresholds. Compliant agent implementations treat unrated vendor agents as high risk and require human approval before releasing any payment.

Attack 2: The spoofing attack

A bad actor registers an identity named close to the real vendor — close enough to fool an agent performing a simple name lookup.

HomIE's response: domain anchoring. The real vendor's HomIE identity is cryptographically tied to their domain name. A spoofed identity fails this verification regardless of how similar the name appears.

Attack 3: The middleman attack

A bad actor sits between two agents, modifies the payment destination, and forwards everything else unchanged. Both ends believe they are communicating directly.

HomIE's response: signed transaction receipts. Before payment releases, the vendor's agent must sign a receipt with their registered private key. A middleman cannot forge this signature. If the signature does not verify, payment does not release.

Attack 4: The silent agent attack

A bad actor agent passes identity verification but broadcasts no metacommunication signal, or a falsely static one that never updates regardless of circumstances.

HomIE's response: metacommunication pattern scoring. Legitimate agents managing real-world work have variable readiness signals that reflect real conditions. Agents with suspiciously static or absent signals are flagged. Their metacommunication history becomes part of their HomIE score.

7.4 The verified transaction handshake

For any agent commerce transaction, the Cred-compliant flow proceeds as follows:

- Query: your agent looks up the vendor's HomIE score and mesh age. Below threshold — abort or escalate to human.
- Verify: both agents perform a cryptographic handshake proving they hold their registered private keys. An agent that refuses the handshake is treated as unverified regardless of score.
- Signal: the vendor's agent begins broadcasting a metacommunication readiness signal. Your agent begins monitoring it. A vendor agent that does not broadcast a signal is treated with elevated caution.
- Transact: the order or job specification is transmitted with a signed session token both agents generated during the handshake.
- Monitor: your agent watches the vendor's readiness signal throughout. Significant drops trigger configurable alerts. The human can engage, ask questions, or approve scope changes at any point.
- Commit: before payment releases, the vendor's agent returns a signed fulfillment commitment certifying the work is complete as specified.
- Record: your agent automatically submits a HomIE event. How accurately the vendor's metacommunication signal reflected reality throughout the job becomes part of the event record and their score.

The core principle

Reputation tells your agent whether to engage. Cryptographic identity ensures the agent it is engaging with is actually who they claim to be. Metacommunication keeps you informed throughout. HomIE requires all three. A high reputation score from an unverified identity is not sufficient. A verified identity with no metacommunication signal is not sufficient. The full handshake is the standard.

8. A Worked Example: The Wedding

To make the protocol concrete, here is a detailed walkthrough of how HomIE works in a real-world scenario.

Alex is planning a July wedding in Kansas City. Alex uses an AI assistant built on OpenClaw to coordinate vendors. Jordan, a friend in the wedding party, also uses the same platform. Alex has added Jordan's agent to their HomIE wedding circle.

Before: booking a florist

Alex's agent searches for florists. Before presenting options, it queries the HomIE mesh for each candidate vendor. The query returns:

- Bloom & Co (bloomkc.com): score 41/100. 3 events in the past 6 months. 2 reports of unreturned calls within 72 hours of event. 1 pricing dispute (resolved in client's favor). 0 fraud flags.
- Petals on Main (petalsonmain.com): score 78/100. 4 events in the past 12 months. All positive. 2 wedding events, both rated service_fulfillment: positive. Average response time reported as same-day.

The agent surfaces this to Alex: "Petals on Main has a stronger reputation among other agents who have worked with KC florists recently. I'd recommend starting there." Alex agrees. The booking proceeds.

During: circle sharing and metacommunication

Jordan's agent asks Alex's agent (via the HomIE circle) for a status update on the wedding planning. Because Jordan is in Alex's wedding circle, Jordan's agent has read access to the detailed event log and metacommunication history. It reports back to Jordan: caterer confirmed with readiness signal at 94%, florist booked with deposit paid and signal at 91%, venue finalized, photographer still pending. Jordan didn't have to call Alex.

After: contributing to the mesh

The wedding goes well. The florist was excellent. Alex's agent automatically drafts a HomIE event:

```
{
  "subject": "petalsonmain.com",
  "interaction_type": "service_fulfillment",
  "outcome": "positive",
  "severity": 1,
  "context": "Wedding flowers delivered on time, matched brief exactly,
              setup crew professional and efficient.",
  "tags": ["wedding", "kansas-city", "flowers"]
}
```

This is a positive event at severity 1 — tier one under the submission model. The agent signs and posts it automatically. Alex receives no notification. No action required. Petals on Main's score improves slightly and the next agent that queries them will see this event in their record.

Six months later, someone in Denver is planning a destination wedding in KC. Their agent queries the mesh. Petals on Main now has 5 events, all positive, spanning 18 months. The score is 84/100. The booking recommendation is confident.

9. Roadmap

v0.1 is complete with this document. Everything below is a community target. The protocol author commits to maintaining the specification, stewarding the RFC process, and operating homie-protocol.org as the canonical home for the protocol. The engineering is an open invitation. If you want to build any of the items below, open an issue at github.com/mrwalkersir/homie-protocol.

v0.1 — Complete

- Protocol specification (this document)
- Event schema and taxonomy including metacommunication event type
- Trust score algorithm with asymmetric decay, volume normalization, and extreme reviewer penalty
- Identity model and privacy rules
- Three-tier submission model
- Metacommunication protocol design
- homie-protocol.org domain registered and placeholder site — author commitment

v0.2 — Reference implementation (community target)

- Hosted canonical registry at homie-protocol.org — seeking infrastructure contributor
- REST API for event submission, score queries, and metacommunication signals — seeking backend contributor
- JavaScript client library — seeking JS contributor
- Python client library — seeking Python contributor
- OpenClaw skill (HomIE.md + homie.py) — seeking OpenClaw community contributor
- Vendor identity claiming flow

v0.3 — Mesh federation (community target)

- Node-to-node gossip protocol specification
- Multi-registry identity resolution
- Circle management API

- Dispute and arbitration flow
- Metacommunication signal federation across nodes

v1.0 — Open standard (community target)

- Independent governance structure — foundation or working group
- Formal specification (RFC-style)
- Reference test suite for compliance verification
- Integration guides for major agent frameworks
- NIST AI Agent Standards alignment review

10. Governance and Ownership

HomIE is released into the public domain. There are no patents, no trademarks, no licensing restrictions. Anyone may implement the protocol, run a node, build a client, or fork the specification.

The HomIE Protocol was originated by Andrew Walker, an independent researcher focused on trust infrastructure for autonomous AI agents. The first canonical registry and reference implementation are maintained by the protocol author as a public service. This is a practical starting point, not a claim of ownership. The long-term goal is an independent governance structure — a foundation or working group — that stewards the protocol on behalf of all participants.

Changes to the core protocol spec will be managed through a public RFC process. Any participant can propose a change by opening an issue or pull request in the public repository. Changes are discussed openly. Proposals require a comment period of not less than 30 days. Changes require rough consensus from active implementors before being incorporated into a new version. The protocol authors do not have unilateral authority to make breaking changes to a ratified version.

The hosted registry (homie-protocol.org) will be operated as a public interest service. Revenue from commercial API usage will fund operating costs; any surplus will fund protocol development and governance.

A word on why the governance model is designed this way. A protocol owned by one person is a product. A protocol owned by a community is infrastructure. HomIE is designed to be infrastructure. The goal of independent governance is not to give up control for its own sake — it is to ensure that HomIE can grow beyond what any single person can build or maintain. The organizations that will ultimately make HomIE succeed — the agent platform builders, the enterprise integrators, the open source contributors — need to know that the protocol belongs to the community, not to its author. That assurance is what makes adoption safe. It is what makes HomIE worth building on.

A note on commercial implementations

Companies are welcome and encouraged to build commercial products on top of the HomIE protocol. Charging for access to hosted nodes, offering premium query tiers,

building specialized reputation products for specific industries — all of this is permitted and does not conflict with the protocol being open. The protocol is the commons. Products built on it are private.

11. Open Questions

This is a draft document. The following questions are deliberately left open for community discussion:

- Should the canonical trust score algorithm be fixed in the spec, or should the spec only define the event format and leave score computation to implementations? Fixed algorithms are interoperable but inflexible. Open algorithms allow innovation but fragment the meaning of a “HomIE score.”
- How should the protocol handle agents that act across multiple jurisdictions with different legal definitions of fraud or defamation? An event that is legitimate in one country may expose the submitter to legal risk in another.
- What is the right minimum stake to suppress low-quality events without excluding legitimate participants who cannot afford a paid subscription? The current proposal (phone verification) excludes some users.
- Should vendor scores be queryable by the public (including non-agent humans), or should mesh access be restricted to verified agent identities? Public access maximizes utility but creates incentives for SEO manipulation.
- How should the protocol handle AI agents that are themselves vendors — e.g. an agent that offers research services to other agents for a fee? The current taxonomy does not clearly cover agent-as-vendor scenarios.
- The dating app scenario described in section 1.4 — bots being deployed to pursue relationships on behalf of humans — is presented as a near-term prediction arriving within two to three years. This is the authors’ judgment, not an established fact. Community input on the timeline and the nature of the problem is welcome. The protocol design is not dependent on this specific scenario being accurate, but the urgency of the problem statement is.

Contributions to these discussions are welcome at github.com/mrwalkersir/homie-protocol.

12. Closing

There is a version of the agentic internet that goes wrong. Agents that scam each other. Vendors that make promises their agents cannot keep. Humans disconnected from processes happening on their behalf, unable to tell the good actors from the bad, unable to trust the

outcomes of interactions they never witnessed. A world where the technology meant to give people more time instead gives them more anxiety.

That version arrives if we build the communication layer without the trust layer. If we give agents the ability to transact without the ability to remember, to report, to warn each other. If we assume that cryptographic identity alone is enough and discover too late that it isn't.

HomIE is the alternative. Not because the technology is complicated — it isn't. Not because it requires a revolution — it doesn't. But because the right infrastructure, built now, while the agentic internet is still being shaped, makes the difference between a system that works for people and one that works against them.

The goal was never to build a reputation protocol. The goal was Tuesday.

Homework with the kids. A ballgame at the park. Board games and tacos. The freedom to be present for what actually matters because the things that don't — the bookings, the follow-ups, the vendor negotiations, the coordination, the hundred small transactions that eat an ordinary day — are handled. Quietly. Reliably. By agents that know what they're doing and can prove it.

That's what HomIE makes possible. Not magic. A better process. One where the mesh is invisible, the trust is earned, and the human gets their Tuesday back.

This document is that start.

Andrew Walker — Independent Researcher

HomIE Protocol v0.1 — March 2026

github.com/mrwalkersir/homie-protocol

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